

Diff-in-diff

Advanced Applied Econometrics

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Slides based on Scott Cunningham and Paul Goldsmith-Pinkham
Thanks also to Renke Schmacker

Differences-in-differences

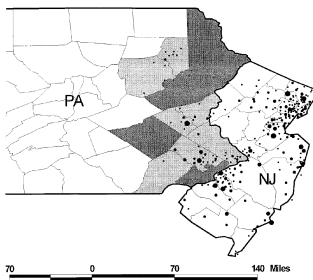
- Goal: estimate effect of a policy
- Naive approach: compare outcomes for treated and untreated groups
- Problem: but policy-adoption is not random $\Rightarrow$ what is the counterfactual?
- Solution: estimate difference-in-difference (DiD)

Differences-in-differences: Card and Krueger (1994)

- Suppose you are interested in the effect of minimum wages on employment (a classic and controversial question in labor economics).
- In a competitive labor market, increases in the minimum wage would move us up a downward sloping labor demand curve → employment would fall

DD: Card and Krueger (1994)

- Card and Krueger (1994) analyzed the effect of a minimum wage increase in New Jersey using a differences-in-differences (DD) methodology
- In February 1992, New Jersey increased the state minimum wage from \$4.25 to \$5.05. Pennsylvania's minimum wage stayed at \$4.25.



- They surveyed about 400 fast food stores both in New Jersey and Pennsylvania before and after the minimum wage increase in New Jersey

DD Strategy

- DD is a version of fixed effects estimation. To see this formally:

Y_{ist}^1 : employment at restaurant i , state s , time t with a high w^{min}

Y_{ist}^0 : employment at restaurant i , state s , time t with a low w^{min}

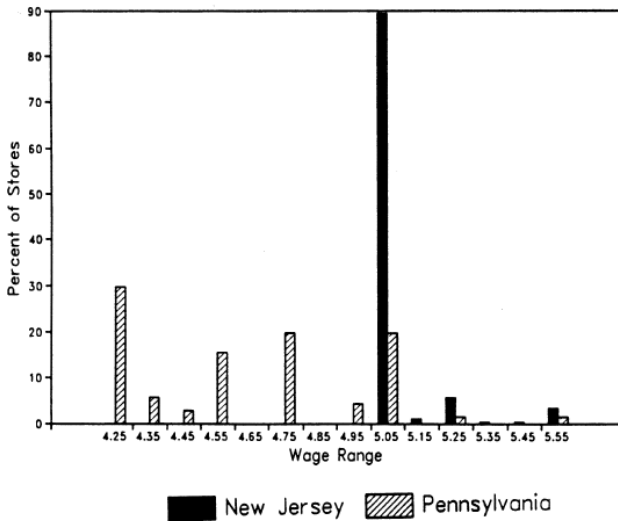
- In practice of course we only see one or the other. We then assume that:

$$E[Y_{its}^0 | s, t] = \gamma_s + \lambda_t$$

- In the absence of a minimum wage change, employment is determined by the sum of a time-invariant state effect, γ_s and a year effect, λ_t that is common across states
- Let D_{st} be a dummy for high-minimum wage states and periods
- Assuming $E[Y_{its}^1 - Y_{its}^0 | s, t] = \delta$ is the treatment effect, observed employment can be written:

$$Y_{its} = \gamma_s + \lambda_t + \delta D_{st} + \varepsilon_{its}$$

November 1992



DD Strategy II

- In New Jersey

- Employment in February is

$$E(Y_{ist}|s = NJ, t = Feb) = \gamma_{NJ} + \lambda_{Feb}$$

- Employment in November is:

$$E(Y_{ist}|s = NJ, t = Nov) = \gamma_{NJ} + \lambda_{Nov} + \delta$$

- Difference between November and February

$$E(Y_{ist}|s = NJ, t = Nov) - E(Y_{ist}|s = NJ, t = Feb) = \lambda_{Nov} - \lambda_{Feb} + \delta$$

- In Pennsylvania

- Employment in February is

$$E(Y_{ist}|s = PA, t = Feb) = \gamma_{PA} + \lambda_{Feb}$$

- Employment in November is:

$$E(Y_{ist}|s = PA, t = Nov) = \gamma_{PA} + \lambda_{Nov}$$

- Difference between November and February

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- Difference between November and February

$$E(Y_{ist}|s = PA, t = Nov) - E(Y_{ist}|s = PA, t = Feb) = \lambda_{Nov} - \lambda_{Feb}$$

DD Strategy III

- The DD strategy amounts to comparing the change in employment in NJ to the change in employment in PA.
- The population DD are:

$$\begin{aligned} & \left(E(Y_{ist}|s = NJ, t = Nov) - E(Y_{ist}|s = NJ, t = Feb) \right) \\ & - \left(E(Y_{ist}|s = PA, t = Nov) - E(Y_{ist}|s = PA, t = Feb) \right) = \delta \end{aligned}$$

- This is estimated using the sample analog of the population means

Variable	Stores by state		
	PA (i)	NJ (ii)	Difference, NJ – PA (iii)
1. FTE employment before, all available observations	23.33 (1.35)	20.44 (0.51)	– 2.89 (1.44)
2. FTE employment after, all available observations	21.17 (0.94)	21.03 (0.52)	– 0.14 (1.07)
3. Change in mean FTE employment	– 2.16 (1.25)	0.59 (0.54)	2.76 (1.36)

Surprisingly, employment *rose* in NJ relative to PA after the minimum wage change

Regression DD

- We can estimate the DD estimator in a regression framework
- Advantages:
 - It's easy to calculate the standard errors
 - We can control for other variables which can make the parallel trend assumption more credible and may reduce the residual variance (lead to smaller standard errors)
 - It's easy to include multiple periods
 - We can study treatments with different treatment intensity. (e.g., varying increases in the minimum wage for different states)
- The typical regression model we estimate is

$$\text{Outcome}_{it} = \beta_1 + \beta_2 \text{Treat}_i + \beta_3 \text{Post}_t + \beta_4 (\text{Treat} \times \text{Post})_{it} + \varepsilon_{it}$$

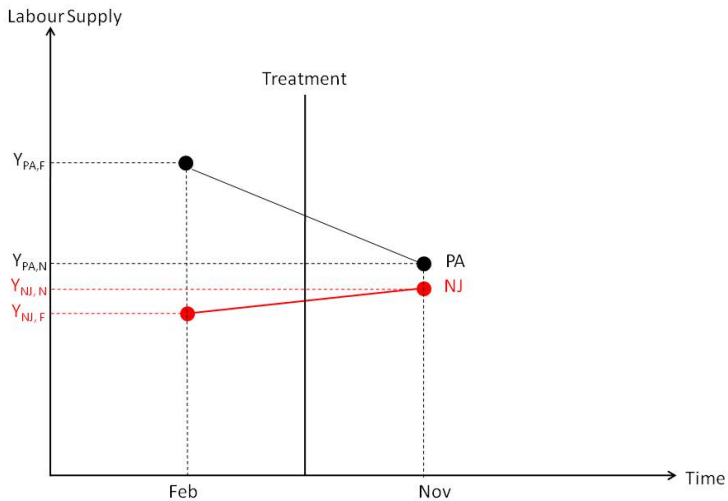
where Treat is a dummy if the observation is in the treatment group and Post is a post treatment dummy

Regression DD - Card and Krueger

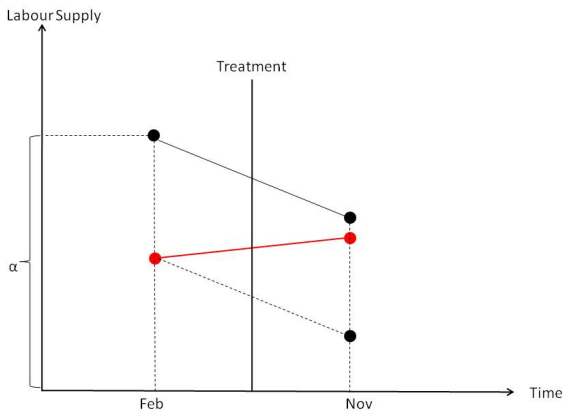
- In the Card and Krueger case, the equivalent regression would be:

$$Y_{its} = \alpha + \gamma NJ_s + \lambda d_t + \delta(NJ \times d)_{st} + \varepsilon_{its}$$

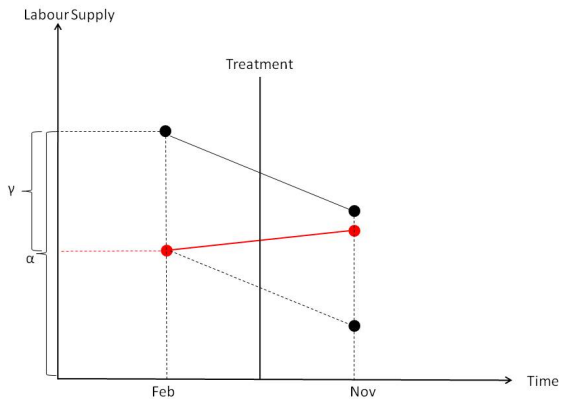
- NJ is a dummy equal to 1 if the observation is from NJ
- d is a dummy equal to 1 if the observation is from November (the post period)
- This equation takes the following values
 - PA Pre: α
 - PA Post: $\alpha + \lambda$
 - NJ Pre: $\alpha + \gamma$
 - NJ Post: $\alpha + \gamma + \lambda + \delta$
- DD estimate: $(NJ \text{ Post} - NJ \text{ Pre}) - (PA \text{ Post} - PA \text{ Pre}) = \delta$
- DD estimates the **ATT**



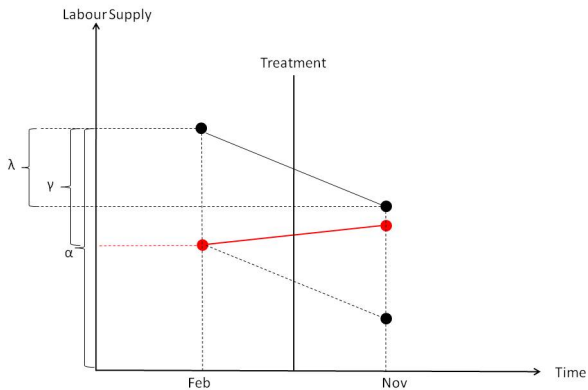
$$Y_{ist} = \alpha + \gamma NJ_s + \lambda d_t + \delta(NJ \times d)_{st} + \varepsilon_{ist}$$



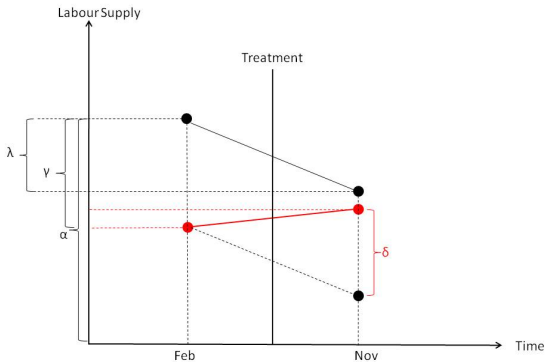
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Assumptions of DD strategy

- Parallel trends:
 - TWFE estimates:

$$\begin{aligned}\beta^{DiD} &= E(Y_{i2}(1) - Y_{i1}(0) | D_{i2} = 1) - E(Y_{i2}(0) - Y_{i1}(0) | D_{i2} = 0) \\ &= E(Y_{i2}(1) - Y_{i1}(0) | D_{i2} = 1) - E(Y_{i2}(0) - Y_{i1}(0) | D_{i2} = 1) \\ &\quad + E(Y_{i2}(0) - Y_{i1}(0) | D_{i2} = 1) - E(Y_{i2}(0) - Y_{i1}(0) | D_{i2} = 0)\end{aligned}$$

- where $E(Y_{i2}(1) - Y_{i2}(0) | D_{i2} = 1) = ATT$ under the parallel trends assumption:

$$E(Y_{i2}(0) - Y_{i1}(0) | D_{i2} = 1) = E(Y_{i2}(0) - Y_{i1}(0) | D_{i2} = 0)$$

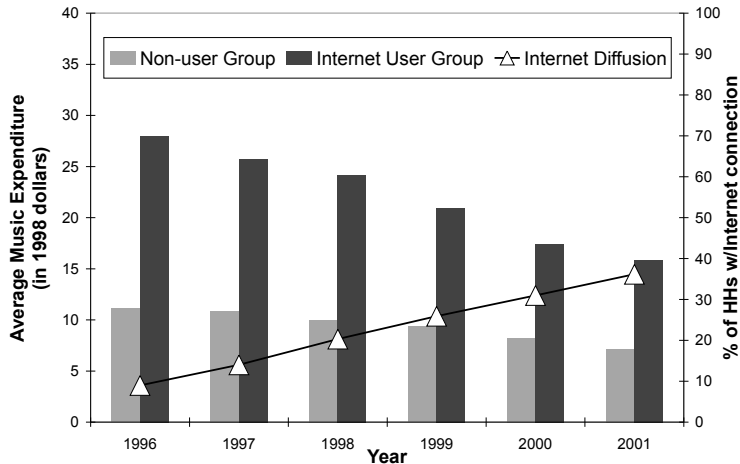
- Stable unit treatment value assumption (SUTVA)
- Fixed sample composition

Threats to validity – compositional differences

- One of the risks of a repeated cross-section (and unbalanced panel) is that the composition of the sample may have changed between the pre and post period
- Hong (2011) uses repeated cross-sectional data from the Consumer Expenditure Survey (CEX) containing music expenditure and internet use for a random sample of households
- Study exploits the emergence of Napster (first file sharing software widely used by Internet users) in June 1999 as a natural experiment
- Study compares internet users and internet non-users before and after emergence of Napster

Compositional differences?

Figure 1: Internet Diffusion and Average Quarterly Music Expenditure in the CEX



Threats to validity – compositional differences

Table 1: Descriptive Statistics for Internet User and Non-user Groups^a

Year	1997		1998		1999	
	Internet User	Non-user	Internet User	Non-user	Internet User	Non-user
Average Expenditure						
Recorded Music	\$25.73	\$10.90	\$24.18	\$9.97	\$20.92	\$9.37
Entertainment	\$195.03	\$96.71	\$193.38	\$84.92	\$182.42	\$80.19
Zero Expenditure						
Recorded Music	.56	.79	.60	.80	.64	.81
Entertainment	.08	.32	.09	.35	.14	.39
Demographics						
Age	40.2	49.0	42.3	49.0	44.1	49.4
Income	\$52,887	\$30,459	\$51,995	\$28,169	\$49,970	\$26,649
High School Grad.	.18	.31	.17	.32	.21	.32
Some College	.37	.28	.35	.27	.34	.27
College Grad.	.43	.21	.45	.21	.42	.20
Manager	.16	.08	.16	.08	.14	.08

Diffusion of the Internet changes samples (e.g., younger music fans are early adopters)

Threats to validity – SUTVA

- Treatment status of any individual does not affect the potential outcomes of other units (non-interference)
 - Violation: Spillovers and general equilibrium effects
- No change in the composition of the treatment and control group due to the treatment
 - Violation: For example selective migration into area where treatment is introduced

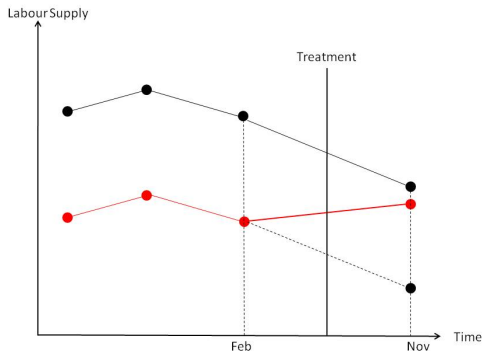
Threats to validity – Non-parallel trends

- Often policymakers will select the treatment and controls based on pre-existing differences in outcomes – practically guaranteeing the parallel trends assumption will be violated.
- “Ashenfelter dip”
 - Named after Orley Ashenfelter, labor economist at Princeton
 - Participants in job trainings program often experience a “dip” in earnings just prior to entering the program
 - Since wages have a natural tendency to mean reversion, comparing wages of participants and non-participants using DD leads to an upward biased estimate of the program effect.
- Regional targeting. NGOs may target villages that appear most promising, or worse off, which is a form of selection bias and violates parallel trends

Key assumption of any DD strategy: Parallel trends

- The key assumption for any DD strategy is that the outcome in treatment and control group would follow the same time trend in the absence of the treatment
- This doesn't mean that they have to have the same mean of the outcome
- Parallel trends are difficult to verify because technically one of the parallel trends is an unobserved counterfactual
- But one often will check using pre-treatment data to show that the trends are the same

Key assumption of any DD strategy: Parallel trends



Even if pre-trends are the same one still has to worry about other policies changing at the same time (omitted variable bias)

Regression DD Including Leads and Lags

- Suppose we have multiple periods before and after the treatment
 - Including leads into the DD model is an easy way to analyze pre-treatment trends
 - Lags can be included to analyze whether the treatment effect changes over time after assignment
- The estimated regression would be:

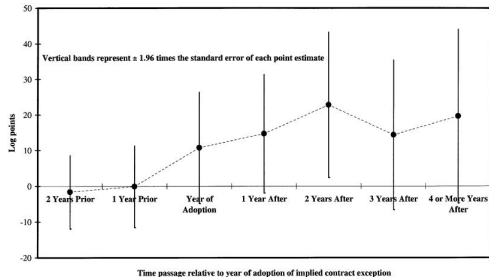
$$Y_{its} = \gamma_s + \lambda_t + \sum_{\tau=-q}^{-1} \gamma_{\tau} D_{s\tau} + \sum_{\tau=0}^m \delta_{\tau} D_{s\tau} + x_{ist} + \varepsilon_{ist}$$

- Treatment occurs in year 0
- Includes q leads or anticipatory effects
- Includes m lags or post treatment effects
- One of the coefficients is unidentified (TE relative to this period)

Study including leads and lags – Autor (2003)

- Autor (2003) includes both leads and lags in a DD model analyzing the effect of increased employment protection on the firm's use of temporary help workers
- In the US employers can usually hire and fire workers at will
- Some states courts have made some exceptions to this employment at will rule and have thus increased employment protection
- The standard thing to do is normalize the adoption year to 0
- Autor then analyzes the effect of these exemptions on the use of temporary help workers.

Results



- The leads are very close to 0. → no evidence for anticipatory effects (good news for the parallel trends assumption).
- The lags show the effect increases during the first years of the treatment and then remains relatively constant.

Testing parallel pre-trends

- Diverging pre-trends can falsify the parallel trends assumption
- Showing parallel pre-trends lends support to the validity of the design (worth doing!)
- However, recent literature pointed out issues with testing pre-trends
 - Parallel trends in what?
 - Pre-testing issues

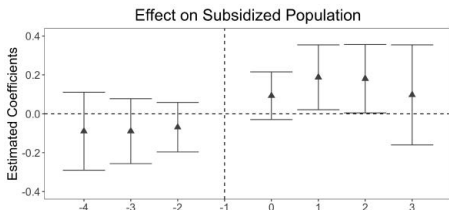
Parallel trends in what?

- How is the outcome specified: in logs or levels?
- If you have parallel pre-trends in logs, they are unlikely to hold in levels and vice versa
- OLS does not have the invariance property that, e.g., quantile regression has
- Roth and Sant'Anna (2021):

“Our results suggest that researchers who wish to point-identify the ATT should justify one of the following: (i) why treatment is as-if randomly assigned, (ii) why the chosen functional form is correct at the exclusion of others, or (iii) a method for inferring the entire counterfactual distribution of untreated potential outcomes.”

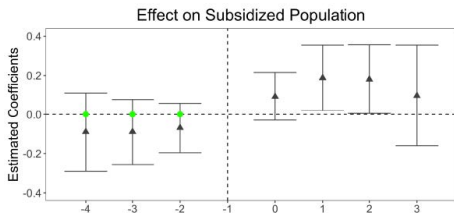
Pre-testing issues (Roth 2020)

- “Event study” visualization of pre-trends
- Zero is in the 95% CIs, so we can't reject parallel trends
- But a sizable pre-trend is in the 95% CI as well
- With low power, we cannot rule out an important alternative hypothesis
- With highly precise pre-trends, the problem is less relevant



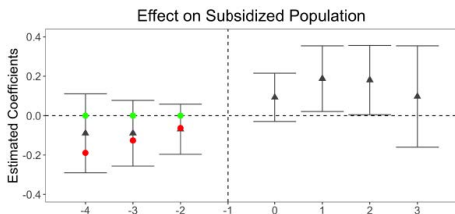
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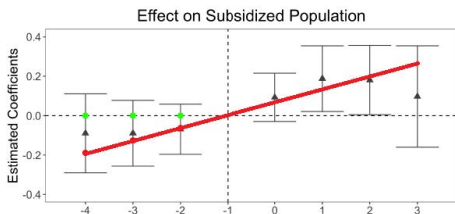
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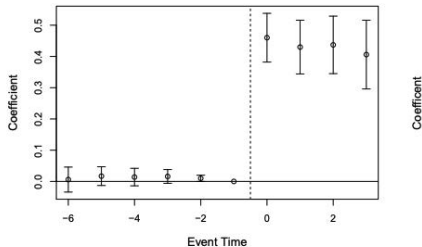
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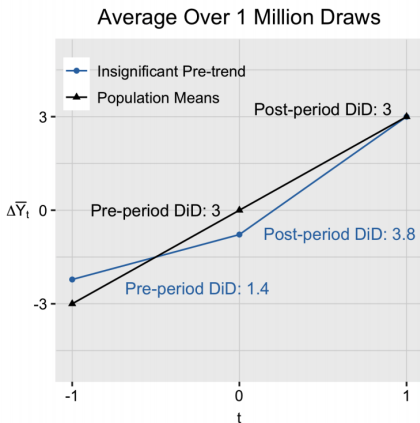
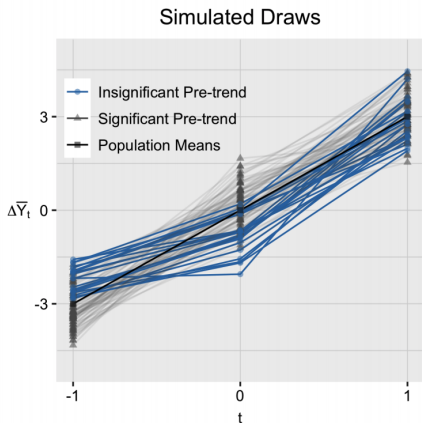
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Outcome: Medicaid Eligibility

Pre-testing issues (Roth 2020)

- By selecting on pre-trends that “pass”, will tend to choose baseline realizations that satisfy pre-trends, but induce *bias* in the effect



Pre-testing issues (Roth 2020)

- First, don't panic. Examining pre-trend is still important diagnostic
- Important to realize that selecting your design based on pre-trend is *constructing* your counterfactual
 - Pre-tests will cause you to potentially contaminate your design
- Suggested solution from Roth (2020): incorporate robustness to pre-trends into your analysis. Rambachan and Roth (2020) present results on testing sensitivity of DiD results to pre-trends
 - Brief intuition follows

Rambachan and Roth (2020) suggestion

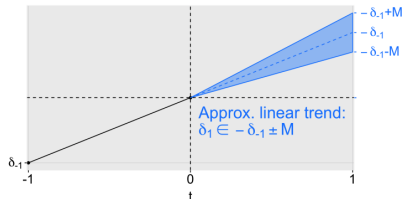
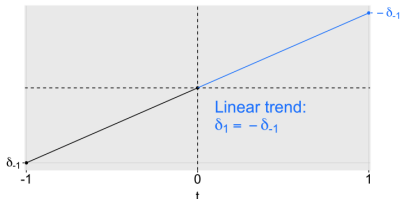
- Intuitive proposed solution for robustness. Note the post and pre

$$\mathbb{E}[\hat{\beta}_1] = \tau_{ATT} + \underbrace{\mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 1] - \mathbb{E}[Y_{i,1}(0) - Y_{i,0}(0) \mid D_i = 0]}_{\text{Post-period differential trend} =: \delta_1},$$

$$\mathbb{E}[\hat{\beta}_{-1}] = \underbrace{\mathbb{E}[Y_{i,-1}(0) - Y_{i,0}(0) \mid D_i = 1] - \mathbb{E}[Y_{i,-1}(0) - Y_{i,0}(0) \mid D_i = 0]}_{\text{Pre-period differential trend} =: \delta_{-1}}.$$

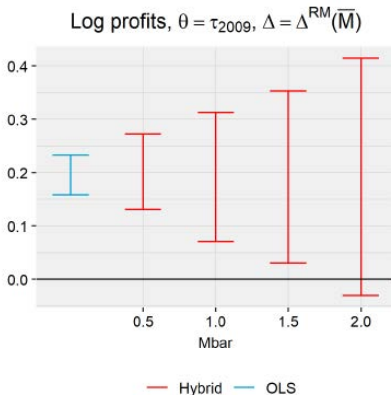
effects:

- parallel trends assumes these δ are zero. But pre-trends may not be zero.
 - R&R say: we can use the info from our pre-trends to bound post-trend
 - Use a *smoothness* assumption, M , on the second derivative. E.g.
- simple case:



Rambachan and Roth (2020) suggestion

Using the honestdid package, we can find the largest M such that the 95% CI excludes zero while requiring $|\delta_1^+| \leq M|\delta_1^-|$



We can rule out a null effect unless we allow for violations of parallel trends that are twice as large than the max in the pre-period!

Checks for DD Design

- Very common for readers and others to request a variety of “robustness checks” from a DD design
- Think of these as along the same lines as the leads and lags we already discussed
 - Falsification test using data for prior periods (already discussed)
 - Falsification test using data for alternative control group
 - Falsification test using alternative “placebo” outcome that should not be affected by the treatment

Alternative control group – DDD

TABLE 3—DDD ESTIMATES OF THE IMPACT OF STATE MANDATES
ON HOURLY WAGES

Location/year	Before law change	After law change	Time difference for location
A. Treatment Individuals: Married Women, 20–40 Years Old:			
Experimental states	1.547 (0.012) [1,400]	1.513 (0.012) [1,496]	–0.034 (0.017)
Nonexperimental states	1.369 (0.010) [1,480]	1.397 (0.010) [1,640]	0.028 (0.014)
Location difference at a point in time:	0.178 (0.016)	0.116 (0.015)	
Difference-in-difference:	–0.062 (0.022)		
B. Control Group: Over 40 and Single Males 20–40:			
Experimental states	1.759 (0.007) [5,624]	1.748 (0.007) [5,407]	–0.011 (0.010)
Nonexperimental states	1.630 (0.007) [4,959]	1.627 (0.007) [4,928]	–0.003 (0.010)
Location difference at a point in time:	0.129 (0.010)	0.121 (0.010)	
Difference-in-difference:	–0.008: (0.014)		
DDD:	–0.054 (0.026)		

DDD in Regression

$$W_{ijt} = \alpha + \beta_1 X_{ijt} + \beta_2 \tau_t + \beta_3 \delta_j + \beta_4 D_i + \beta_5 (\delta \times \tau)_{jt} \\ + \beta_6 (\tau \times D)_{ti} + \beta_7 (\delta \times D)_{ij} + \beta_8 (\delta \times \tau \times D)_{ijt} + \varepsilon_{ijt}$$

- The DDD estimate is the difference between the DD of interest and a placebo DD (which is supposed to be zero)
- If the placebo DD is non-zero, it might be difficult to convince the reviewer that the DDD removed all the bias
- If the placebo DD is zero, then DD and DDD give the same results but DD is preferable because standard errors are smaller for DD than DDD

Standard errors in DD strategies

- If you are using panel data, outcomes and the treatment tend to be severely autocorrelated
- Using robust standard errors will lead to overrejection of the H_0 due to downward biased standard errors (see Bertrand, Duflo, Mullainathan 2004)
- **If you have enough clusters, you must cluster on the unit of policy implementation.**
 - If the policy is implemented at the industry level, you should cluster at the industry and not at the firm level
- In the Card and Krueger study (1 treated, 1 control state), clustering at the state level is infeasible

Standard errors in DD strategies – few clusters

- Often, we have less than 42 clusters (MHE) in our diff-in-diff
- Solutions to calculate standard errors with few clusters have been proposed but this is an active field of research
- Highly context specific and often involves making trade-offs between (strong) assumptions
 - Donald and Lang (2007) two-step procedure
 - Conley and Taber (2010) if you have many control groups and few treated groups
 - (Wild) Cluster bootstrap (e.g., Cameron, Gelbach, Miller 2008)
 - Recent work by Andreas Hagemann

Cases of DiD

- 1 treatment timing, Binary treatment, 2 periods
 - Card and Krueger (AER, 1994)
- 1 treatment timing, Binary treatment, T periods
 - Yagan (AER, 2015)
- Staggered treatment timing, Binary treatment
 - Bailey and Goodman-Bacon (AER, 2015)